

1. Create a Person parent class with name and age. Create a Student child class with roll_no and marks. Use super() to initialize the parent attributes.
2. Create a Vehicle parent class with a start() method. Create Car and Bike child classes and override the start() method with different behavior.
3. Create an Employee parent class with name and salary. Create a Manager child class with an additional department attribute. Display all details using constructors.
4. Create a BankAccount parent class with account_number and balance. Create a SavingsAccount child class with an additional interest_rate. Use super() in the child constructor.
5. Create a Person parent class with a display() method. Create a Student child class that overrides display() but also calls the parent display() using super().
6. Create a parent class Company with a company_name attribute. Create two child classes Developer and Tester, each with their own additional attributes and methods.

7. **Bank Account**

Create a BankAccount class with a private attribute __balance.

Create get_balance() to view the balance.

Create set_balance() to update the balance only if the amount is valid.

Create a withdraw() method with proper balance validation.

8. **Student Marks**

Create a Student class with private attributes __name and __marks.

Create getter and setter methods for marks.

Allow marks only between 0 and 100.

Create a child class ExamStudent that inherits from Student

9. **Constructor + Private Data**

Create a User class with private __password.

Initialize it through the constructor.

Create get_password() and set_password().

Setter should update the password only if it contains at least 6 characters.

10. **Multilevel Inheritance + Getter/Setter**

Person → private age with getter/setter.

Employee → salary.

Manager → department.

Use super() at each level